

Latency Report

Pipeline: raw STT -> LLM -> TTS | Mode: LLM | Streaming: off | Sample size: 7 interactions

This baseline measures the current non-streaming pipeline. Audio is recorded in the browser, sent to the backend for transcription, passed to the LLM, converted to speech, and then sent back to the browser for playback.

Executive findings

- The largest user-visible delay is not the local tool chain. The local tool functions are almost free because they run in-process over local sample data.
- Tool turns in LLM mode pay for two model steps: one model pass to decide the tool call, and a second model pass to turn the tool result into a spoken answer. The dashboard currently reports those together as model response time.
- The biggest raw bottleneck in this sample is TTS completion. The backend receives the first audio chunk earlier, but the current app waits for the full MP3 before the browser can play it.
- End-of-turn detection is stable at about 0.68s per turn. That is expected for a conservative VAD/turn-detection setting, but it is still meaningful latency before STT even starts.
- Because this sample has only seven interactions, p95 should be treated as a directional worst-case signal rather than a statistically stable production number.

Summary metrics

Metric	What it means	Median	P95
End-of-turn detection	Delay after the user stops speaking before the turn is accepted.	0.68s	0.69s
Speech-to-text	Time to transcribe the captured user audio.	1.14s	2.58s
LLM first token	Time until the model starts producing output.	1.28s	1.63s
LLM total	Total model time. On tool turns this combines planning plus post-tool response.	2.01s	2.78s
Local tool chain	Time spent executing local backend tools and mock lookups.	0.00s	0.01s
TTS first audio	Time until the backend receives the first audio bytes from TTS.	2.19s	4.54s
TTS complete	Time until the full speech audio is generated.	6.50s	24.11s
Browser delivery	Client-side delivery and playback-start overhead after backend processing.	0.04s	0.20s
End-to-heard	Total delay from user finishing speaking to the browser starting the response.	10.37s	27.46s

Cleaned interaction table

All timings below are shown in seconds. A dash means that stage did not apply to the turn.

#	User said	Tool?	End turn	STT	LLM	Tool chain	TTS first	TTS full	End-to-heard
1	Hello	No	0.68s	3.13s	1.37s	-	2.02s	3.19s	8.50s
2	What can you do for me?	No	0.68s	1.14s	2.01s	-	1.73s	6.50s	10.37s
3	I wanted an update on my order.	No	0.69s	1.30s	1.37s	-	2.19s	6.19s	9.58s
4	I'm Arjun Mehta.	Yes	0.68s	1.15s	2.81s	0.01s	1.04s	5.46s	10.35s
5	Your order ID is?	Yes	0.69s	0.95s	2.71s	0.00s	3.99s	15.28s	19.66s
6	It's 2204.	Yes	0.69s	1.04s	2.71s	0.00s	3.01s	15.01s	19.50s
7	No just this much only.	No	0.68s	1.00s	1.17s	-	4.78s	27.90s	30.79s

How to read the baseline

In this LLM-mode baseline, tool execution itself is not the latency problem. The model response column includes every LLM pass required by the turn, so tool turns include both tool planning and post-tool verbalization. The TTS columns reveal the current architectural limitation: the backend can see first audio earlier than the browser, but playback still waits for complete synthesis. The next optimization should therefore stream audio to the browser as soon as usable TTS chunks arrive.